

Macroscopic-microscopic analysis of deformation energy, Q_{20} electric quadrupole moments, and equilibrium shapes in even-even barium and cerium isotopes

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ABSTRACT

This study investigates the electric quadrupole moments (Q_{20}) and deformation energies of even-even nuclei within the mass range $114 \leq A \leq 140$, focusing on Barium and Cerium isotopes. A macroscopic-microscopic approach is employed, combining the liquid drop model for macroscopic energy with Strutinsky shell corrections and BCS (*Bardeen-Cooper-Schrieffer*) pairing correlations for microscopic refinements. The shell corrections are derived by solving a Schrödinger equation with a deformed Woods-Saxon mean field. Numerical solutions, performed using an optimized FORTRAN program, identify equilibrium shapes by locating energy minima in deformation energy contour plots. From these equilibrium shapes, single-particle wave functions and the BCS approximation are used to calculate electric quadrupole moments. Theoretical results align closely with experimental data, confirming the robustness of the proposed model. This study advances our understanding of nuclear deformation and structure, particularly in isotopes far from spherical symmetry.

1. Introduction

The shape and deformation of atomic nuclei are fundamental topics in nuclear physics, as they deeply influence the structure, stability, and behavior of nuclei. Historically, nuclei were assumed to be spherical. However, experimental measurements, especially of electric quadrupole moments (Q_{20}), have revealed significant deviations from sphericity in many nuclei [1]. Such deformations are particularly pronounced in nuclei far from magic numbers, where shell effects and pairing interactions strongly impact their properties

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[2].

The study of nuclear deformation is critical for understanding various phenomena, including nuclear fission, collective rotational and vibrational bands, and electromagnetic transition rates [3]. Deformation is typically quantified using parameters like the quadrupole deformation (β) and the electric quadrupole moment (Q_{20}), which provide key insights into the spatial distribution of nuclear charge [4].

To model these deformations and predict nuclear structure properties, macroscopic-microscopic (Mac-Mic) methods have proven particularly effective. These approaches combine global macroscopic trends with local quantum corrections, enabling accurate predictions across wide mass regions. Recent studies have further refined this framework, especially in the treatment of shell and pairing effects. For instance, Bhagwat et al. [5] proposed a refined Mac-Mic model based on the Wigner-Kirkwood expansion applied to a deformed Woods-Saxon potential. Their model, tested on 551 even-even nuclei, achieved an rms deviation of 610 keV, comparable to those of FRDM (*Finite Range Droplet Model*) and LSD (*Lublin-Strasbourg Drop model*) models. Pomorski et al. [6] studied the stability of superheavy nuclei using a Yukawa-folded potential combined with Strutinsky shell corrections and BCS (*Bardeen-Cooper-Schrieffer approximation*) pairing. Their results showed good agreement with experimental data for binding energies and fission barriers, validating the approach in regions far from stability. Kowal and Skalski [7] demonstrated that Mac-Mic models offer strong predictive capabilities for fission barrier heights and energy surfaces. Their study emphasized the flexibility of this method compared to self-consistent mean-field models. Verriere and Mumpower [8] introduced technical improvements to the Strutinsky smoothing and pairing correlations (*Lipkin-Nogami scheme*), reducing spurious effects and enhancing the resolution of potential energy surfaces for fission applications.

Building on these advances, this study employs a macroscopic-microscopic approach, combining the liquid drop model for macroscopic energy with microscopic corrections based on the Strutinsky method and the Bardeen-Cooper-Schrieffer (BCS) pairing approximation [9]. Additionally, the Schrödinger equation is solved using a Woods-Saxon deformed potential to account for microscopic interactions with higher accuracy [10]. This study focuses on even-even isotopes of Barium and Cerium in the mass range $114 \leq A \leq 140$, a transitional region where theoretical modeling remains comparatively limited despite the availability of experimental data. By focusing on this mass region, the study aims to provide a detailed analysis of equilibrium shapes, deformation energies, and electric quadrupole moments, contributing both high-resolution theoretical predictions and critical comparisons with available experimental values.

The primary objectives of this study are:

- **Determine equilibrium nuclear shapes:** Identify the equilibrium configurations of nuclei by analyzing minima in deformation energy surfaces, computed using the Strutinsky method [11].
- **Evaluate Q_{20} electric quadrupole moments:** Calculate these moments using single-particle wave functions and pairing corrections to understand their relationship with nuclear deformation [12].
- **Compare theoretical predictions with experimental data:** Validate the accuracy and predictive power of the proposed model by comparing theoretical results with experimental values [13].

By integrating macroscopic and microscopic methods and leveraging an optimized *FORTRAN* program, this work provides detailed insights into the structure, deformation, and stability of nuclei across this isotopic range. Furthermore, this study bridges gaps in existing nuclear deformation models and demonstrates the critical role of advanced computational approaches in modern nuclear physics [14].

2. Microscopic model and details of calculations

2.1. Schrödinger equation

The particle Schrödinger equation, assuming imposed symmetries, incorporates kinetic energy, potential energy, spin-orbit interaction, and Coulomb energy for protons [15]. It is expressed as:

$$\hat{H}|\psi_i\rangle = \varepsilon_i|\psi_i\rangle, \quad (\text{Eq. 1})$$

where $|\psi_i\rangle$ and ε_i denote the nucleon's eigenfunctions and eigenenergies, respectively. These quantities are obtained by solving the Schrödinger equation numerically in a deformed harmonic oscillator basis with axial symmetry. Gauss quadrature methods are used to compute the matrix elements, and the resulting matrix is diagonalized to extract eigenvalues and eigenfunctions. Cartesian coordinates are preferred for triaxial deformation scenarios.

The nucleon Hamiltonian is given by:

$$\hat{H} = T + V + V^{SO} + e\phi^c, \quad (\text{Eq. 2})$$

where T is the kinetic energy, V the potential energy, V^{SO} the spin-orbit interaction, and $e\phi^c$ the Coulomb energy. The matrix elements of $\hat{H}$ can be written as:

$$\langle n'_x n'_y n'_z \Sigma' | \hat{H} | n_x n_y n_z \Sigma \rangle = \langle n'_x n'_y n'_z \Sigma' | (T + V + V^{SO} + e\phi^c) | n_x n_y n_z \Sigma \rangle. \quad (\text{Eq. 3})$$

The Woods–Saxon potential is employed to describe the mean field experienced by nucleons due to its ability to represent the realistic distribution of nuclear matter. The potential is mathematically expressed as [16]:

$$V(r) = \frac{-V_0}{1 + \exp\left(\frac{r - R_V}{a_V}\right)}, \quad (\text{Eq. 4})$$

where $R_V = r_0 A^{-\frac{1}{3}}$ with $r_0 = 1.2 \text{ fm}$, representing the nuclear radius; V_0 is the potential depth, and a_V is the surface diffuseness parameter.

For deformed nuclei, the potential is generalized as:

$$V(r) = \frac{-V_0}{1 + \exp\left(\frac{R_V L_V}{a_V}\right)}, \quad (\text{Eq. 5})$$

here, L_V (*quasi-radius*), accounts for the deformation of the nucleus [17].

The Woods–Saxon potential is particularly well-suited for modeling nuclear deformations due to its incorporation of a surface diffuseness parameter [18], which accurately captures the gradual transition at the nuclear surface. This property ensures a realistic representation of nuclear matter distribution, especially for nuclei with large deformations [19]. Moreover, its mathematical formulation allows for straightforward extensions to include axial and triaxial deformations, making it ideal for the study of non-spherical nuclei [20]. The inclusion of spin-orbit coupling further enhances its ability to replicate key nuclear structure features observed experimentally [21]. These characteristics make the Woods–Saxon potential a robust choice for calculating mean fields in the context of deformed even-even nuclei.

2.2. Pairing corrections (BCS)

The pairing correlations are taken into account using the Bardeen-Cooper-Schrieffer (BCS) approximation in the potential energy. The Fermi level λ and the gap parameter Δ are determined by solving the gap equations:

$$\Delta = \frac{G}{2} \sum_{k=0}^{N_p} \frac{\Delta}{\sqrt{(\epsilon_k - \lambda)^2 + \Delta^2}}, \quad (\text{Eq. 6})$$

$$N \text{ or } Z = \sum_{k=0}^{N_p} \left(1 - \frac{(\epsilon_k - \lambda)}{\sqrt{(\epsilon_k - \lambda)^2 + \Delta^2}} \right), \quad (\text{Eq. 7})$$

where G is the pairing constant, ϵ_k represents the eigenvalues of the microscopic Hamiltonian h , and N_p is the number of nucleon pairs considered (half above and half below the sharp Fermi level). The empirical average gap is approximated by:

$$\bar{\Delta} = 12 / \sqrt{A} \text{ MeV}. \quad (\text{Eq. 8})$$

The pairing strength G is connected to the smooth density of states $\bar{g}(\lambda)$ by the relation:

$$\frac{1}{G} \approx \bar{g}(\lambda) \ln \left(\frac{N_p}{\bar{g}(\lambda) \Delta} \right). \quad (\text{Eq. 9})$$

The quasi-particle energy associated with the k -th orbital is given by:

$$E_k = \sqrt{(\epsilon_k - \lambda)^2 + \Delta^2}. \quad (\text{Eq. 10})$$

Once the pairing gap parameter Δ and the Fermi level λ are determined, the occupation probabilities of nucleon pairs are calculated as:

$$\nu_k^2 = \frac{1}{2} \left[1 - \frac{\epsilon_k - \lambda}{(\epsilon_k - \lambda)^2 + \Delta^2} \right], \quad k = 1, \dots, N_p. \quad (\text{Eq. 11})$$

This comprehensive approach ensures that pairing corrections are accurately accounted for, significantly improving the predictive capability of the macroscopic-microscopic method for deformed nuclei.

2.3. The potential energy of deformation

The deformation energy is one of the most significant quantities in nuclear structure studies. It is calculated using the macroscopic-microscopic method [22], where the shell correction is handled using the Strutinsky procedure [23]. The energy of nucleus deformation is expressed as:

$$E_{def}(N, Z, \beta, \gamma) = E_{LD}(N, Z, \beta, \gamma) - E_{LD}(N, Z, 0, 0) = E_s(N, Z, \beta, \gamma) + E_c(N, Z, \beta, \gamma) - E_s(N, Z, 0, 0) - E_c(N, Z, 0, 0), \quad (\text{Eq. 12})$$

here, $E_{LD}(\beta, \gamma)$ represents the deformation-dependent component of the liquid-drop binding energy [24], E_s and E_c are the surface and Coulomb energy contributions, respectively. The liquid-drop energy is expressed as:

$$E_{LD}(N, Z, \beta, \gamma) = \frac{3}{5} \frac{e^2}{r_0} \frac{Z^2}{A^{1/3}} \left[\frac{A}{2Z^2} \xi(B_s - 1) + (B_c - 1) \right], \quad (\text{Eq. 13})$$

where, B_s and B_c are dimensionless surface and Coulomb energy factors relative to the spherical shape. ξ is the reduced fissility parameter, determined experimentally [25], and $r_0 = 1.275 \text{ fm}$, $e^2 = 1.4399764 \text{ MeV}$.

The reduced fissility parameter is defined as:

$$\zeta = 52.8(1 - 2.84I^2), \quad (\text{Eq. 14})$$

where I is the asymmetry parameter:

$$I = \frac{(N - Z)}{(N + Z)}. \quad (\text{Eq. 15})$$

The shell correction $\delta E(N, Z)$ is evaluated using the Strutinsky method separately for neutrons and protons as:

$$\delta E_{1,2}(N \text{ or } Z, \beta, \gamma) = \sum_i^{NorZ} \varepsilon_i - \overline{\sum_i \varepsilon_i}, \quad (\text{Eq. 16})$$

here, ε_i is the energy of a particle (*individual energy*), and the sums $\sum_i^{NorZ} \varepsilon_i - \overline{\sum_i \varepsilon_i}$ are related to ρ and $\bar{\rho}$, respectively. ρ represents the Hartree-Fock (HF) density matrix, while $\bar{\rho}$ is a smoothed quantity that does not include shell effects. The averaged energy are defined through the average density of states $\bar{g}_{M,\gamma}(\varepsilon)$ by

$$\overline{\sum_i \varepsilon_i} = \int_{-\infty}^{\bar{\lambda}} \varepsilon \bar{g}_{M,\gamma}(\varepsilon) d\varepsilon, \quad (\text{Eq. 17})$$

where $\bar{g}_{M,\gamma}(\varepsilon)$ represents the uniform level density calculated using the Strutinsky method, and $\bar{\lambda}$ is the corresponding Fermi level.

The pairing correction is calculated separately for protons and neutrons as [26]:

$$\delta P_{1,2}(N \text{ or } Z, \beta, \gamma) = P - \bar{P}, \quad (\text{Eq. 18})$$

where P represents the pure pairing energy and $\bar{P}$ is the averaged pairing energy (without shell effects). The pairing energy and the averaged pairing energy are expressed as:

$$P = \sum_{k=1}^{N_p} 2v_k^2 \varepsilon_k - \frac{\Delta^2}{G} - \sum_{k=1}^{N_p/2} 2\varepsilon_k, \quad (\text{Eq. 19})$$

$$\bar{P} = -\frac{1}{2} g_{str}(\bar{\lambda}) \bar{\Delta}^2 = -\frac{1}{2} \bar{g}(\bar{\lambda}) \bar{\Delta}^2, \quad (\text{Eq. 20})$$

where g_{str} represents the level density calculated using the Strutinsky method, v_k is the occupation probability of quasi-particle pairs, and G denotes the pairing constant. The eigenvalues of the microscopic Hamiltonian $\hat{H}$, corresponding to independent particles, are represented by ε_k . The parameter N_p refers to the number of particle pairs considered in the calculation, with $N_p/2$ being the number of particles on either side of the Fermi level in the independent particle model (IPM), corresponding to the bound states between the Fermi level and the zero level.

The deformation energy is calculated as:

$$E_{def}(N, Z, \beta, \gamma) = E_{LD}(N, Z, \beta, \gamma) + \delta E_{shell}(N, Z, \beta, \gamma) + \delta E_{pairing}(N, Z, \beta, \gamma) \quad (\text{Eq. 21})$$

Expanding this expression:

$$E_{def}(N, Z, \beta, \gamma) = E_{LD}(N, Z, \beta, \gamma) + \delta E_1(Z, \beta, \gamma) + \delta E_2(N, \beta, \gamma) + \delta P_1(Z, \beta, \gamma) + \delta P_2(Z, \beta, \gamma) \quad (\text{Eq. 22})$$

3. The electric quadrupole moment operator of the nucleus

The non-normalised quadrupole moment operator Q_{20} , is defined as:

$$Q_{20} = \sum_{i=1}^Z e \sqrt{\frac{16\pi}{5}} r_i^2 Y_{20}(\theta_i, \varphi_i), \quad (\text{Eq. 23})$$

where e is the charge of the proton, $(r_i, \theta_i, \varphi_i)$ are the spherical coordinates of the i th proton, and $Y_{lm}(\theta, \varphi)$ is a spherical harmonic, which is independent of φ when $l = 0$.

The quadrupole moment can also be expressed in Cartesian coordinates as [27]:

$$Q_{20} = \sum_{i=1}^Z e(3z_i^2 - r_i^2), \quad (\text{Eq. 24})$$

with

$$r_i^2 = x_i^2 + y_i^2 + z_i^2. \quad (\text{Eq. 25})$$

For a nucleus in the state ψ , the mean value of the quadrupole moment is given by:

$$Q_{20}(\psi) = \langle \psi | Q_{20} | \psi \rangle = \sum_{i=1}^Z \int \int \int e(3z_i^2 - r_i^2) |\psi(r_1, \dots, r_z)|^2 d\tau. \quad (\text{Eq. 26})$$

3.1. The spectroscopic quadrupole moments

In the wave function describing the nucleus, the collective degrees of freedom are characterized by three quantum numbers: I , which represents the total angular momentum; M , the projection of M on the OZ -axis; and K , the projection of I on the OZ' -axis, which is intrinsic to the nucleus.

The spectroscopic quadrupole moment (or measured quadrupole moment) corresponds to the mean value of the quadrupole moment operator when $M = I$. It is expressed as:

$$Q = \langle \psi | Q_{20} | \psi \rangle_{M=I}, \quad (\text{Eq. 27})$$

This average value can be written as:

$$Q = (II20II)(IK20IK)Q_{20}, \quad (\text{Eq. 28})$$

where the terms in parentheses are Clebsch-Gordan coefficients. Here, Q_{20} is the intrinsic quadrupole moment, calculated in the intrinsic reference frame, and depends only on the intrinsic degrees of freedom.

Using the properties of Clebsch-Gordan coefficients, the spectroscopic quadrupole moment is given by:

$$Q = \frac{3K^2 - I(I+1)}{(I+1)(2I+3)} Q_{20}. \quad (\text{Eq. 29})$$

For the ground state of a rotational band, where $I = K$, the spectroscopic quadrupole moment simplifies to:

$$Q = \frac{I(2I+1)}{(I+1)(2I+3)} Q_{20}. \quad (\text{Eq. 30})$$

3.2. The intrinsic quadrupole moments

The intrinsic quadrupole moment can be determined more easily through reduced transition probabilities. It is expressed as:

$$Q_{20}(eb) = \sqrt{\frac{16\pi}{5}} B(E2; 0^+ \rightarrow 2^+), \quad (\text{Eq. 31})$$

where $B(E2; 0^+ \rightarrow 2^+)$ represents the probability of electromagnetic transitions. Experimental measurements of nuclear deformation often involve evaluating the quadrupole moment classically (using a liquid-drop model) under the assumption of small quadrupole deformation β and axial symmetry [28]. The resulting expression is:

$$Q_{20} \approx ZR_{ch}^2 \frac{3}{\sqrt{5\pi}} \left(\beta + \frac{2}{7} \sqrt{5\pi} \beta^2 \right), \quad (\text{Eq. 32})$$

where Z is the total charge of the nucleus and R_{ch} is the charge radius.

For a homogeneous liquid drop of axially symmetrical ellipsoidal shape, the quadrupole deformation β is approximately:

$$\beta = \frac{4}{3} \sqrt{\frac{\pi}{5}} \left(\frac{\Delta R}{R} \right), \quad (\text{Eq. 33})$$

where $\Delta R = c - a$, with a, c are the semi-axes of the ellipsoid. The effective nuclear surface is generally an ellipsoid characterized by

three semi-axes, described by:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \quad (\text{Eq. 34})$$

Due to volume conservation, the effective nuclear surface is parameterized by only two quantities: the elongation β and the axial asymmetry γ , commonly referred to as the Bohr parameters [29]. For the Woods–Saxon potential, several studies have introduced and applied the so-called universal parameters to describe nuclear deformations across a wide range of isotopes. These parameters, detailed in Table 1, have been extensively developed and validated in works such as [30–33], and provide a reliable framework for modeling the shape-dependent features of nuclear structure.

The experimental values of the electric quadrupole moment are derived from the transition probabilities $B(E2; 0^+ \rightarrow 2^+)$ using the following relationship [34]:

$$Q_{20}^{\text{exp}} = \sqrt{\frac{16\pi}{5}} B(E2)_{\text{exp}} = \beta_2^{\text{exp}} \frac{3Z}{\sqrt{5\pi}} R^2. \quad (\text{Eq. 35})$$

The nuclear radius R is expressed as:

$$R = r_0 A^{\frac{1}{3}}, \quad r_0 = 1.2 \text{ fm}. \quad (\text{Eq. 36})$$

3.3. The quadrupole moment in the independent particle model (IPM)

The quantum quadrupole moment is defined by:

$$Q_{20} = \frac{1}{e} \sum \int \int \int \psi^* q_i (3Z_i^2 - r_i^2) \psi d^3r, \quad (\text{Eq. 37})$$

where ψ represents the nuclear wave function, and the summation covers all the nucleons, each with a charge q_i . r_i^2 is the square of the radial distance for the i th proton.

In the independent particle model (IPM), matching correlations are introduced using the BCS method. With this approach, the quantum quadrupole moment Q_{20} becomes:

$$Q_{20} = \sum_{i=1}^{N_p} 2\nu_i^2 \langle \psi_i | \hat{Q}(i) | \psi_i \rangle, \quad (\text{Eq. 38})$$

where ν_i^2 represents the occupation probability of the state $|\psi_i\rangle$, which is the wave function of the Schrödinger equation in the IPM. The parameter N_p corresponds to the number of quasi-particle pairs.

The sign of Q_{20} determines the nuclear shape (Fig. 1):

- If $Q_{20} < 0$, the nucleus has an oblate shape (*flattened ellipsoid*).
- If $Q_{20} = 0$, the nucleus is spherical.
- If $Q_{20} > 0$, the nucleus has a prolate shape (*elongated ellipsoid*).

The calculation procedure follows three main steps. First, the deformation energy is computed for the nucleus. Next, the absolute minimum of the deformation energy is determined for each nucleus. Finally, at this minimum, the quadrupole moment is calculated using the individual wave functions, incorporating the BCS approximation in all calculations.

4. Results and discussion

4.1. Typical examples of deformation energy

The deformation energy surfaces calculated using the Strutinsky method for Barium $^{118}_{56}\text{Ba}$ and Cerium $^{122}_{58}\text{Ce}$ are shown in Figs. 2

Table 1

The parameters of the Woods–Saxon potential.

Parameters	Neutrons	Protons
V_0 (Potential depth) [MeV]	$-49.6[1 - 0.86(N - Z)/(N + Z)]$	$-49.6[1 + 0.86(N - Z)/(N + Z)]$
a_V (Potential diffuseness) [fm]	0.70 fm	0.70 fm
R_V (Potential radius) [fm]	$1.347 A^{1/3}$	$1.275 A^{1/3}$
κ (Spin-orbit coupling)	$35.0 V_0 (\hbar/2m_n c)^2 \approx 35.0 V_0 \times 0.01104$	$35.0 V_0 (\hbar/2m_n c)^2 \approx 35.0 V_0 \times 0.01104$
a_{so} (Spin-orbit diffuseness) [fm]	0.70 fm	0.70 fm
R_{so} (Spin-orbit radius) [fm]	$1.310 A^{1/3}$	$1.20 A^{1/3}$

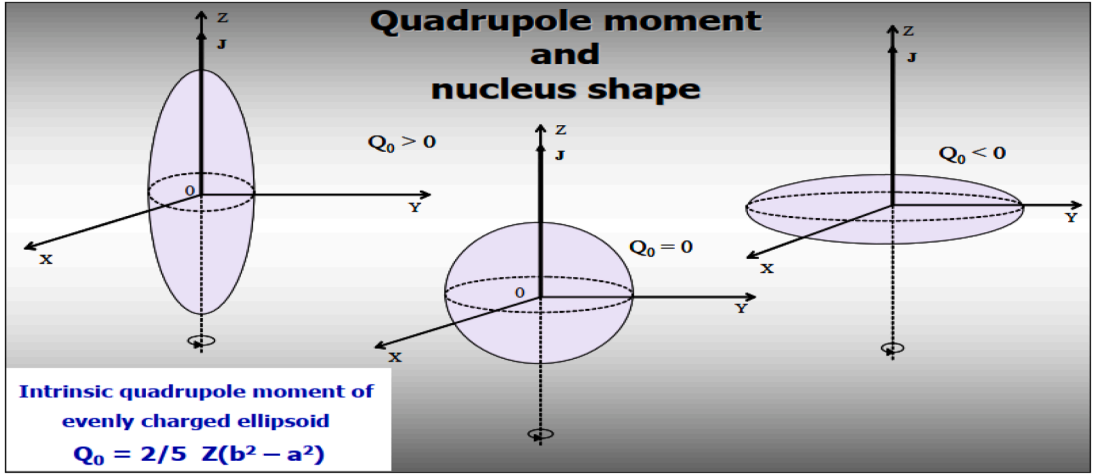


Fig. 1. Representation of nuclear shapes based on the quadrupole moment.

and 3, respectively. These results illustrate the variation of the deformation energy E_{def} (MeV) as a function of the deformation parameters β and γ , with $0 \leq \beta \leq 0.7$ and $0^\circ \leq \gamma \leq 60^\circ$. The calculations were performed with step sizes $\Delta\beta = 0.1$ and $\Delta\gamma = 10^\circ$, here the spherical configuration corresponds to $\beta = 0$.

4.2. Equilibrium deformations and electric quadrupole moments

4.2.1. The potential energy surfaces

The deformation energy curves for $^{118}_{56}\text{Ba}$ and $^{122}_{58}\text{Ce}$ are typical examples (Figs. 2 and 3). The minima of these curves correspond to the equilibrium shapes of the nuclei. The quadrupole deformation parameter β is limited to 0.7, and the physically interesting region for the minima is $0 < \beta < 0.25$.

For $^{118}_{56}\text{Ba}$, the equilibrium deformation is approximately $(\beta_{st.eq}, \gamma_{st.eq}) \approx (0.25, 0^\circ)$, and for $^{122}_{58}\text{Ce}$, it is $(\beta_{st.eq}, \gamma_{st.eq}) \approx (0.28, 0^\circ)$. These values indicate the static equilibrium deformation of the nuclei. The deformation energy, defined as the energy difference between the spherical configuration ($\beta = 0$) and the equilibrium deformation (β_{eq}), provides insight into the stability of the nuclei. For $^{118}_{56}\text{Ba}$, the deformation energy is approximately $E_{def} = (5.4266 - 1.4542) \approx 3.9724$ MeV, suggesting that this nucleus is relatively "soft" or only slightly deformed. For $^{122}_{58}\text{Ce}$, the deformation energy is $E_{def} = (7.8052 - 1.2994) \approx 6.5058$ MeV, indicating a more significantly deformed nucleus.

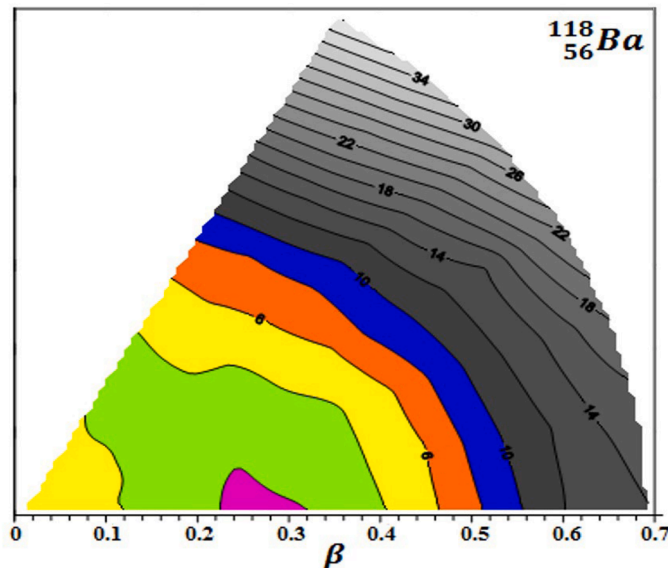


Fig. 2. Deformation energy contour plot for $^{118}_{56}\text{Ba}$, calculated using the Strutinsky method in the (β, γ) -plane. Contour levels are expressed in MeV.

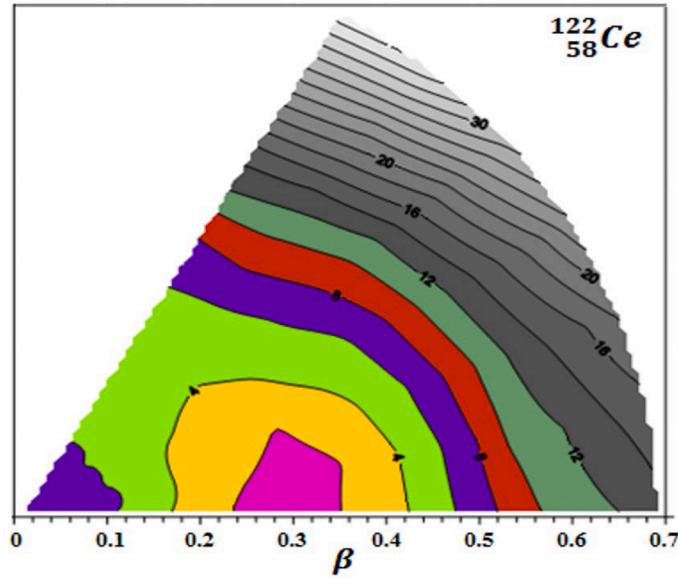


Fig. 3. Deformation energy contour plot for $^{122}_{58}\text{Ce}$, calculated using the Strutinsky method in the (β, γ) -plane. Contour levels are expressed in MeV.

The results for the equilibrium deformations (β, γ) and deformation energies E_{def} for the Barium and Cerium isotopes are summarized in Tables 2 and 3. Experimental deformations, determined from transition probabilities $B(E2; 0^+ \rightarrow 2^+)$ [35], are compared to theoretical values, along with results from Arseniev et al. [13].

4.2.2. Electrical quadrupole moments

The electric quadrupole moments provide further evidence of nuclear deformation. These moments are calculated using various experimental techniques and are derived from reduced transition probabilities $B(E2; 0^+ \rightarrow 2^+)$. Tables 4 and 5 summarize the theoretical axial quadrupole moments $Q_{20}(\text{theo})$, the experimental values $Q_{20}(\text{exp})$ obtained from transition probabilities, and the quadrupole moments measured directly. The results are compared with those from Arseniev et al. [13]. The theoretical results indicate that the isotopes of Barium in this region generally have small deformation energies, whereas Cerium isotopes exhibit more significant deformation. Most equilibrium deformations correspond to $\gamma = 0^\circ$, indicating predominantly prolate shapes with axial symmetry.

4.3. Key findings

- The Barium isotopes in this region are characterized by relatively small deformation energies, while Cerium isotopes exhibit significant deformation. Equilibrium deformations for most isotopes yield $\gamma = 0^\circ$, indicating prolate axial symmetry.
- Significant deformations are observed near $N \approx 66$, midway between the magic numbers $N = 50$ and $N = 82$. Cerium isotopes ($Z = 58$) exhibit pronounced deformations, strongly correlated with larger quadrupole moment values. The isotopes

Table 2

Static equilibrium deformation parameters and deformation energy values for Barium isotopes.

A, N	β_{eq}	$\gamma_{\text{eq}} [^\circ]$	E_{def} ($\beta = 0$) [MeV]	E_{def} (at β_{eq}) [MeV]	E_{def} [MeV]	$\beta_{\text{exp}}(B(E2) \uparrow)$	$\beta_{2\text{eq}}$ (Arseniev)	E_{def} (Arseniev) E_{def} [MeV]
$^{114}\text{Ba}_{58}$	0.21	0°	3.6446	0.5709	3.0737	—	—	—
$^{116}\text{Ba}_{60}$	0.29	0°	6.1852	1.5015	4.6837	—	0.33	2.8
$^{118}\text{Ba}_{62}$	0.25	0°	5.4266	1.4542	3.9724	—	0.32	3.2
$^{120}\text{Ba}_{64}$	0.29	0°	5.878	1.8538	4.0242	—	0.31	3.3
$^{122}\text{Ba}_{66}$	0.34	0°	6.6709	1.1526	5.5183	0.354	0.30	3.2
$^{124}\text{Ba}_{68}$	0.35	0°	5.5161	2.2931	3.223	0.302	0.27	2.8
$^{126}\text{Ba}_{70}$	0.28	0°	4.6258	2.4347	2.1911	0.285	0.23	2.2
$^{128}\text{Ba}_{72}$	0.21	20°	3.2946	1.9038	1.3908	0.238	0.20	1.6
$^{130}\text{Ba}_{74}$	0.16	20°	2.9279	1.0244	1.9035	0.23	0.16	0.9
$^{132}\text{Ba}_{76}$	0.17	20°	0.8497	0.3338	0.5159	0.186	—	—
$^{134}\text{Ba}_{78}$	0.14	20°	−1.4955	−1.6719	0.1764	0.1636	—	—
$^{136}\text{Ba}_{80}$	0.10	40°	−2.7732	−3.6986	0.9254	0.1242	—	—
$^{138}\text{Ba}_{82}$	0.00	0°	−6.3574	−6.3574	0.00	0.0925	—	—

Table 3

Static equilibrium qeformation parameters and deformation energy values for Cerium isotopes.

A, N	β_{eq}	γ_{eq} [°]	E_{def} ($\beta = 0$) [MeV]	E_{def} (at β_{eq}) [MeV]	E_{def} [MeV]	β_{exp} (B(E2) ↑)	β_{2eq} (Arseniev)	E_{def} (Arseniev) E_{def} [MeV]
$^{116}\text{Ce}_{58}$	0.28	0°	5.703	0.7192	4.9838	–	–	–
$^{118}\text{Ce}_{60}$	0.28	0°	5.7792	1.2973	4.4819	–	–	–
$^{120}\text{Ce}_{62}$	0.29	0°	7.4195	0.9554	6.4641	–	–	–
$^{122}\text{Ce}_{64}$	0.28	0°	7.8052	1.2994	6.5058	–	0.33	5.1
$^{124}\text{Ce}_{66}$	0.30	0°	8.1505	1.6293	6.5212	0.302	0.32	5
$^{126}\text{Ce}_{68}$	0.30	0°	7.8	1.7749	6.0251	0.325	0.31	4.5
$^{128}\text{Ce}_{70}$	0.25	0°	6.5545	1.9843	4.5702	0.29	0.27	3.6
$^{130}\text{Ce}_{72}$	0.24	10°	5.8067	2.5475	3.2592	0.257	0.23	2.6
$^{132}\text{Ce}_{74}$	0.22	0°	4.2666	2.1106	2.156	0.257	0.19	1.7
$^{134}\text{Ce}_{76}$	0.18	30°	1.9661	1.4077	0.5584	0.194	0.16	0.7
$^{136}\text{Ce}_{78}$	0.26	30°	0.3795	–1.3569	1.7364	0.17	–	–
$^{138}\text{Ce}_{80}$	0.05	30°	–2.595	–2.8057	0.2107	0.1259	–	–
$^{140}\text{Ce}_{82}$	0.06	40°	–4.8598	–4.9888	0.129	0.1012	–	–

Table 4

Theoretical and experimental quadrupole moments for Barium isotopes.

A, N	$Q_{20(\text{theo})}$ [barn]	$Q_{20}^{\text{exp}} \beta(\text{B(E2)} \uparrow)$	$Q_{20}^{\text{exp}} \beta(\text{Q})$ directly	$Q_{20}(\beta_2)$
$^{114}\text{Ba}_{58}$	3.6036	–	–	–
$^{116}\text{Ba}_{60}$	4.7763	–	–	5.5
$^{118}\text{Ba}_{62}$	4.1899	–	–	5.3
$^{120}\text{Ba}_{64}$	4.775	–	–	5.1
$^{122}\text{Ba}_{66}$	5.5556	5.3152	–	4.9
$^{124}\text{Ba}_{68}$	5.7175	4.5838	–	4.6
$^{126}\text{Ba}_{70}$	4.6246	4.3722	–	3.8
$^{128}\text{Ba}_{72}$	3.3979	3.6897	–	3.2
$^{130}\text{Ba}_{74}$	2.6719	3.6027	–0.3, –0.86, –0.1, –1	2.6
$^{132}\text{Ba}_{76}$	2.8238	2.9433	–	–
$^{134}\text{Ba}_{78}$	2.3664	2.6149	0.21, –0.32, 0.09, –0.2, –0.13, –0.34	–
$^{136}\text{Ba}_{80}$	1.3817	2.0049	0.25, 0.01, 0.07, –0.19	–
$^{138}\text{Ba}_{82}$	–0.0010	1.5077	–0.14, 0.08	–

Table 5

Theoretical and experimental quadrupole moments for Cerium isotopes.

A, N	$Q_{20(\text{theo})}$ [barn]	$Q_{20}^{\text{exp}} \beta(\text{B(E2)} \uparrow)$	$Q_{20}^{\text{exp}} \beta(\text{Q})$ directly	$Q_{20}(\beta_2)$
$^{116}\text{Ce}_{58}$	4,9985	–	–	–
$^{118}\text{Ce}_{60}$	4,9996	–	–	–
$^{120}\text{Ce}_{62}$	5,1403	–	–	–
$^{122}\text{Ce}_{64}$	4,9999	–	–	5.9
$^{124}\text{Ce}_{66}$	5,2812	4,7475	–	5.7
$^{126}\text{Ce}_{68}$	5,2117	5,1639	–	5.6
$^{128}\text{Ce}_{70}$	4,5709	4,6564	–	5.0
$^{130}\text{Ce}_{72}$	4,3603	4,1694	–	4.2
$^{132}\text{Ce}_{74}$	4,1223	4,2121	–	3.4
$^{134}\text{Ce}_{76}$	2,9923	3,2116	–	2.9
$^{136}\text{Ce}_{78}$	4,0786	2,8422	–	–
$^{138}\text{Ce}_{80}$	0.9173	2,1255	–	–
$^{140}\text{Ce}_{82}$	0.9217	1,7249	–	–

$^{114}\text{Ba}_{56}$, $^{116}\text{Ba}_{56}$, $^{118}\text{Ba}_{56}$, $^{120}\text{Ba}_{56}$, $^{122}\text{Ba}_{56}$, $^{124}\text{Ba}_{56}$ and $^{116}\text{Ce}_{58}$, $^{118}\text{Ce}_{58}$, $^{120}\text{Ce}_{58}$, $^{122}\text{Ce}_{58}$, $^{124}\text{Ce}_{58}$, $^{126}\text{Ce}_{58}$, $^{128}\text{Ce}_{58}$, $^{130}\text{Ce}_{58}$ are characterized as deformed, with large equilibrium strains β_{eq} and significant deformation energies (E_{def}).

- Nuclei near $N = 82$ (e.g. $^{118}\text{Ba}_{56}$ and $^{122}\text{Ce}_{58}$) exhibit quasi-spherical shapes, consistent with shell effects from the magic number $N = 82$. Cross-combinations such as $N = 82$, $Z = 56$ and $N = 82$, $Z = 58$ lead to ambiguous cases due to competing shell effects from both protons and neutrons. However, the magic number $N = 82$ exerts a dominant influence on the neutron shell, stabilizing these nuclei in quasi-spherical configurations.
- A strong correlation between the quadrupole deformation parameter β and the electric quadrupole moment Q_{20} is observed. This demonstrates that Q_{20} can be reliably deduced from the deformation parameter β , as the quadrupole moments are largely independent of shell effects.

- The results are consistent with those of Arseniev et al. [13]. Our calculations approximate experimental quadrupole moments with good precision, demonstrating the reliability of the macroscopic-microscopic approach. These findings highlight the critical role of shell effects and deformation energy in shaping nuclear properties, providing valuable insights into regions far from spherical symmetry.

5. Conclusion

This study demonstrated the effectiveness of the macroscopic-microscopic approach, which combines the liquid drop model with shell and pairing corrections, for analyzing nuclear deformations. The shell corrections, derived from a realistic microscopic model incorporating a Woods–Saxon-type mean field, spin-orbit interaction, and a Coulomb field for protons, were calculated following Strutinsky's prescription. Pairing interactions were treated using the BCS approximation. The deformation characteristics of nuclei were found to depend primarily on the proximity of neutron or proton numbers to magic numbers, with nuclei far from magic numbers exhibiting significant deformations, while those near magic numbers remained spherical. The nuclei analyzed showed minimal dependence on the γ -degree of freedom, classifying them as "gamma soft" nuclei. Most equilibrium deformations corresponded to $\gamma = 0^\circ$, indicating a predominantly prolate axial symmetry. The largest deformations were observed near $N \approx 66$, a region midway between the magic numbers $N = 50$ and $N = 82$, for both Barium and Cerium isotopes. Furthermore, the electric quadrupole moments were found to be regular and strongly correlated with the deformation parameter β , confirming the robustness of the theoretical model in capturing deformation-related nuclear properties. The theoretical quadrupole moments closely aligned with experimental values, often matching or exceeding the accuracy of other related methods. These results underscore the reliability of the macroscopic-microscopic method and its potential for advancing our understanding of nuclear structure and deformation.

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CRediT authorship contribution statement

Hadj Mouloudj: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Fethi Redjem:** Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Oussama Zeggai:** Visualization, Validation, Supervision, Methodology, Investigation. **Moussaab Belarbi:** Writing – review & editing, Visualization, Validation, Supervision, Investigation. **Youssef Belgaid:** Visualization, Investigation, Formal analysis, Data curation. **Abdelkader Ghalem:** Visualization, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] S. Leoni, B. Fornal, A. Barcoo, Y. Tsunoda, T. Otsuka, Multifaceted character of shape coexistence phenomena in atomic nuclei, *Progr. Particl. Nucl. Phys.* 139 (2024) 104119, <https://doi.org/10.1016/j.pnpnp.2024.104119>.
- [2] V.M. Strutinsky, Shell effects in nuclear masses and deformation energies, *Nucl. Phys. A* 95 (2) (1967) 420–442, [https://doi.org/10.1016/0375-9474\(67\)90510-6](https://doi.org/10.1016/0375-9474(67)90510-6).
- [3] A. Hosseinnazhad, H. Sabri, Origin of regularity in different nuclei from the viewpoint of the effective quadrupole deformation and triaxiality, *Phys. Script.* 99 (6) (2024), <https://doi.org/10.1088/1402-4896/ad49df>.
- [4] Q.B. Chen, N. Kaiser, U.-G. Meißner, J. Meng, Static quadrupole moments of nuclear chiral doublet bands, *Phys. Lett. B* 807 (2020) 135568, <https://doi.org/10.1016/j.physletb.2020.135568>.
- [5] A. Bhagwat, M. Centelles, X. Viñas, et al., Mic–Mac model based on the Wigner–Kirkwood method, *Eur. Phys. J. A* 59 (299) (2023), <https://doi.org/10.1140/epja/s10050-023-01209-y>.
- [6] K. Pomorski, A. Dobrowolski, B. Nerlo-Pomorska, et al., On the stability of superheavy nuclei, *Eur. Phys. J. A* 58 (77) (2022), <https://doi.org/10.1140/epja/s10050-022-00737-3>.
- [7] M. Kowal, J. Skalski, The fission barrier of heaviest nuclei from a macroscopic-microscopic perspective, in: I. Tanihata, H. Toki, T. Kajino (Eds.), *Handbook of Nuclear Physics*, Springer, Singapore, 2023, https://doi.org/10.1007/978-981-19-6345-2_85.
- [8] M. Verriere, M.R. Mumpower, Improvements to the macroscopic-microscopic approach of nuclear fission, *Phys. Rev. C* 103 (2021) 034617, <https://doi.org/10.1103/PhysRevC.103.034617>.
- [9] Q.-Z. Chai, J.-L. Wang, X.-H. Liu, Effects of triaxial deformation on the fission barrier in the $Z = 118 - 120$ nuclei, *Commun. Theor. Phys.* 76 (2024) 065301, <https://doi.org/10.1088/1572-9494/ad3b8e>.

- [10] H.F. Zhang, A modified macroscopic-microscopic nuclear mass formula within considering isospin effect and continuum states, Nucl. Phys. A 1049 (2024) 122913, <https://doi.org/10.1016/j.nuclphysa.2024.122913>.
- [11] A. Jain, P. Parab, G. Saxena, et al., Impact of nuclear deformation of parent and daughter nuclei on one proton radioactivity lifetimes, Sci. Rep. 14 (2024) 28368, <https://doi.org/10.1038/s41598-024-79726-z>.
- [12] J. Bonnard, J. Dobaczewski, G. Danneaux, M. Kortelainen, Nuclear DFT electromagnetic moments in heavy deformed open-shell odd nuclei, Phys. Lett. B 843 (2023) 138014, <https://doi.org/10.1016/j.physletb.2023.138014>.
- [13] D.A. Arseniev, A. Sobieczewski, V.G. Soloviev, Ground and excited state deformations in the $50 < Z, N < 82$ region, Nucl. Phys. A 126 (1) (1969) 15–35, [https://doi.org/10.1016/0375-9474\(69\)90397-2](https://doi.org/10.1016/0375-9474(69)90397-2).
- [14] N. Wang, J.M. Chen, M. Liu, Effective nucleus-nucleus potentials for heavy-ion fusion reactions, Nucl. Sci. Tech. 36 (24) (2025), <https://doi.org/10.1007/s41365-024-01625-9>.
- [15] J. Damgaard, H.C. Pauli, V.V. Pashkevich, V.M. Strutinsky, A method for solving the independent-particle Schrödinger equation with a deformed average field, Nucl. Phys. A 135 (2) (1969) 432–444, [https://doi.org/10.1016/0375-9474\(69\)90174-2](https://doi.org/10.1016/0375-9474(69)90174-2).
- [16] Z. Kurmanaliyev, D. Soldatkhan, G. Yergaliuly, et al., The impact of deuteron breakup in the field of ^{90}Zr and ^{116}Sn targets, Braz. J. Phys. 55 (67) (2025), <https://doi.org/10.1007/s13538-025-01699-9>.
- [17] H.C. Pauli, On the shell model and its application to the deformation energy of heavy nuclei, Phys. Rep. 7 (2) (1973) 35–100, [https://doi.org/10.1016/0370-1573\(73\)90022-7](https://doi.org/10.1016/0370-1573(73)90022-7).
- [18] J. Ljungberg, Nuclear Structure from a Novel Beyond-Mean-Field Model: Towards Super-Heavy Nuclei from Densities and Mean-Fields to Wave Functions and Observables, Department of Physics, Lund University, 2024. Retrieved from, <https://portal.research.lu.se/en/publications/nuclear-structure-from-a-novel-beyond-mean-field-model-towards-su>.
- [19] A. Ibraheem, F. Alasmay, B. Alshahrani, M. Karakoc, S. Hamada, M.N. El-Hammamy, Comparative study of alpha scattering from various nuclei at 130 MeV: microscopic potentials and coupled channel calculations, Rev. Mex. Fis. 70 (3 May–Jun) (2024) 031202, <https://doi.org/10.31349/RevMexFis.70.031202>, 1.
- [20] B. Mohammed-Azizi, D.E. Medjadi, Single particle calculations for a Woods-Saxon potential with triaxial deformations, and large cartesian oscillator basis, Comput. Phys. Commun 156 (2004) 241–282, [https://doi.org/10.1016/S0010-4655\(03\)00464-8](https://doi.org/10.1016/S0010-4655(03)00464-8).
- [21] M.S. Islam, A.S.B. Tariq, S.K. Joardar, M.A. Uddin, A.K. Basak, Search for the true nature of nucleus-nucleus potentials, J. Sci. Res. 16 (2) (2024) 613–636, <https://doi.org/10.3329/jsr.v16i2.71132>.
- [22] M. Bolsterli, E.O. Fiset, J.R. Nix, J.L. Norton, New calculation of fission barriers for heavy and superheavy nuclei, Phys. Rev. C Nucl. Phys. 5 (3) (1972) 1050–1077, <https://doi.org/10.1103/PhysRevC.5.1050>.
- [23] V.M. Strutinsky, Shells' in deformed nuclei, Nucl. Phys. A 122 (1) (1968) 1–33, [https://doi.org/10.1016/0375-9474\(68\)90699-4](https://doi.org/10.1016/0375-9474(68)90699-4).
- [24] H.C. Pauli, On the shell model and its application to the deformation energy of heavy nuclei, Phys. Rep. 7 (2) (1973) 35–100, [https://doi.org/10.1016/0370-1573\(73\)90022-7](https://doi.org/10.1016/0370-1573(73)90022-7).
- [25] H. Eslamizadeh, H. Raanaei, Fission dynamics of excited compound nuclei in a wide range of fissile parameters produced in $^{16,18}\text{O} + X$ reactions within a stochastic approach, Phys. Rev. C 107 (2023) 054604, <https://doi.org/10.1103/PhysRevC.107.054604>.
- [26] S.M. Wang, W. Nazarewicz, R.J. Charity, L.G. Sobotka, Nucleon–nucleon correlations in the extreme oxygen isotopes, J. Phys. G: Nucl. Part. Phys. 49 (10LT02) (2022), <https://doi.org/10.1088/1361-6471/ac888f>.
- [27] D. Nagy, P. Reinholdt, P.W.K. Jensen, E.R. Kjellgren, K.M. Ziems, A. Fitzpatrick, S. Knecht, J. Kongsted, S. Coriani, S.P.A. Sauer, Electric field gradient calculations for ice VIII and IX using polarizable embedding: a comparative study on classical computers and quantum simulators, J. Phys. Chem. A 128 (30) (2024) 6305–6315, <https://doi.org/10.1021/acs.jpca.4c02697>.
- [28] H. Mouloudj, B. Mohammed-Azizi, O. Zeggai, A. Ghalem, A.E. Toubal Maamar, Estimation of nuclear mass formulas coefficients using least-squares method based on Gauss-Seidel scheme: a comparative study between three models, East Eur. J. Phys. (4) (2023) 37–47, <https://doi.org/10.26565/2312-4334-2023-4-04>.
- [29] A. Maj, A. Bracco, Nuclear Structure at Finite Temperature and the Electric Dipole Oscillations. Oxford Research Encyclopedia of Physics, 2025. Retrieved from, <https://oxfordre.com/physics/view/10.1093/acrefore/9780190871994.001.0001/acrefore-9780190871994-e-116>.
- [30] W. Nazarewicz, J. Dudek, R. Bengtsson, T. Bengtsson, I. Ragnarsson, Microscopic study of the high-spin behaviour in selected $A \simeq 80$ nuclei, Nucl. Phys. A 435 (2) (1985) 397–447, [https://doi.org/10.1016/0375-9474\(85\)90471-3](https://doi.org/10.1016/0375-9474(85)90471-3).
- [31] B.D. Kern, R.L. Mlekodaj, G.A. Leander, M.O. Kortelahti, E.F. Zganjar, R.A. Braga, R.W. Fink, C.P. Perez, W. Nazarewicz, P.B. Semmes, Transition through triaxial shapes of the light samarium isotopes and the beta decay of $^{136,138,140}\text{Eu}$, Phys. Rev. C 36 (4) (1987) 1514–1521, <https://doi.org/10.1103/PhysRevC.36.1514>.
- [32] J. Dudek, Z. Szymanski, T. Werner, A. Faessler, C. Lima, Description of the high spin states in ^{146}Gd using the optimized Woods-Saxon potential, Phys. Rev. C 26 (1982) 1712, <https://doi.org/10.1103/PhysRevC.26.1712>.
- [33] S. Cwiok, J. Dudek, W. Nazarewicz, J. Skalski, T.R. Werner, Single-particle energies, wave functions, quadrupole moments and g-factors in an axially deformed woods-saxon potential with applications to the two-centre-type nuclear problems, Comput. Phys. Commun 46 (3) (1987) 379–399, [https://doi.org/10.1016/0010-4655\(87\)90093-2](https://doi.org/10.1016/0010-4655(87)90093-2).
- [34] A.H. Ali, N.T. Jarallah, S.V. Bedenko, H.A. Alshammar, Comparison between the theoretical and experimental values of transition rate $B(E2)$, deformation parameters and intrinsic quadrupole moments for some Ti isotopes, Appl. Radiat. Isotop. 206 (2024) 111231, <https://doi.org/10.1016/j.apradiso.2024.111231>.
- [35] National Nuclear Data Center. (n.d.). Evaluated Nuclear Structure Data File (ENSDF). Brookhaven National Laboratory. Retrieved from <https://www.nndc.bnl.gov/be2/index.html>.